

TrES-5b

R-1444

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_130531 · created 2026-08-01T08:46:26+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456443.7722 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.174 +/- 0.051
Transit depth (Rp/Rs)^2	3.03 +/- 1.76 [%]
Orbital Inclination (inc)	81.58 +/- 2.94
Airmass coefficient 1 (a1)	0.836 +/- 0.043
Airmass coefficient 2 (a2)	0.104 +/- 0.025
Scatter in the residuals of the lightcurve fit is	5.2 %
Best Comparison Star	#3 - [150, 23]
Optimal Aperture	2.55
Optimal Annulus	13.22
Transit Duration (day)	0.049 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.174 $\pm$ 0.051 vs 0.142 (deviation 0.38 σ)
Duration fit vs published	0.049 d vs 0.0758 d (deviation 0.55 σ)
Mid-time O-C	3.5 min
Depth SNR	2.1
Residual scatter	5.2 %

Light curve

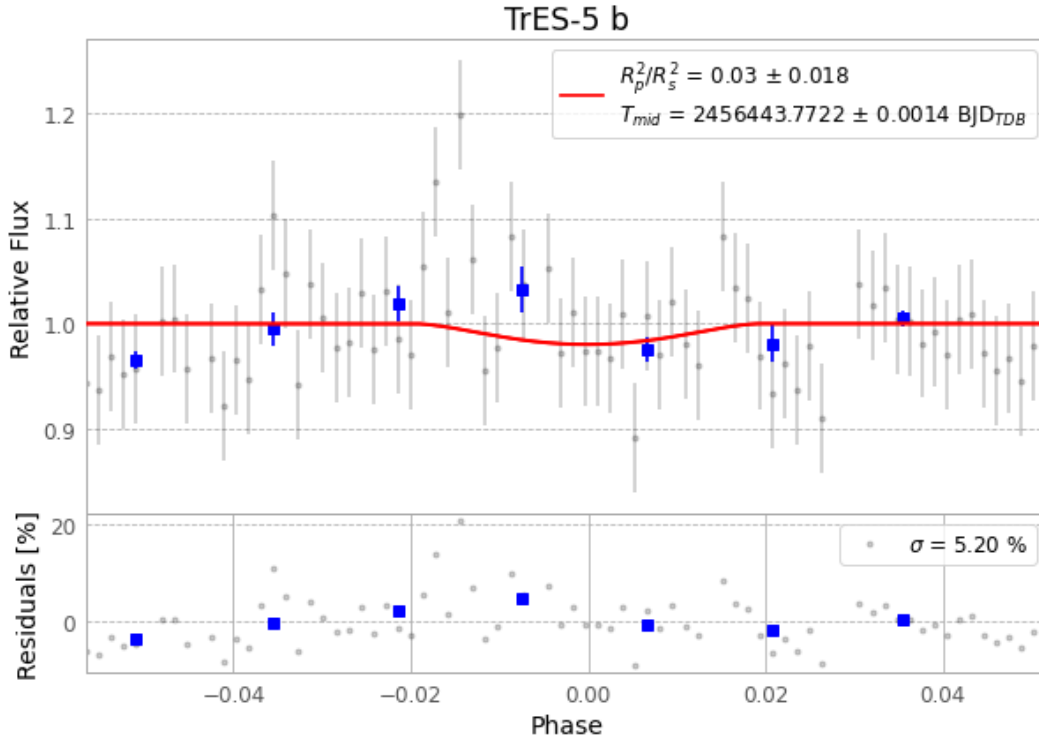


Figure 1 — FinalLightCurve_TrES-5 b_2013-05-30.png (SHA-256 c266920b82339614...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [291, 119], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 89a9a1c4885ca73b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

78 FITS frames (51 MB), TRES-5130531043012.FITS → TRES-5130531082112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-130531.log (6ef467e21940...), FinalLightCurve_TrES-5 b_2013-05-30.png (c266920b8233...), FinalLightCurve_TrES-5 b_2013-05-30.csv (57caaf9c7846...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 08:46Z.